

Transformations of Quadratic Functions

Transformations of Functions

Transformation:

A change made to a figure or a relation such that the figure or the graph of the relation is shifted or changed in shape.

Translations, stretches, and reflections are types of transformations.

The general function:

$$\rightarrow g(x) = af[k(x-d)] + c$$

a transformed function

takes $f(x)$ and performs transformations to it

$f(x)$ parent function you are transforming

Changes to the y-coordinates (vertical changes)

c: vertical translation $g(x) = f(x) + c$

The graph of $g(x) = f(x) + c$ is a vertical translation of the graph of $f(x)$ by c units.

If $c > 0$, the graph shifts **up**

If $c < 0$, the graph shifts **down**

a: vertical stretch/compression $g(x) = af(x)$

The graph of $g(x) = af(x)$ is a vertical stretch or compression of the graph of $f(x)$ by a factor of a .

If $a > 1$ or $a < -1$, **vertical stretch** by a factor of a .

If $-1 < a < 1$, **vertical compression** by a factor of a .

If $a < 0$, **vertical reflection** (reflection over the x-axis)

Note: a vertical stretch or compression means that distance from the x-axis of each point of the parent function changes by a factor of a .

Note: for a vertical reflection, the point (x, y) becomes point $(x, -y)$

Changes to the x-coordinates (horizontal changes)

d: horizontal translation $g(x) = f(x - d)$

The graph of $g(x) = f(x - d)$ is a horizontal translation of the graph of $f(x)$ by d units.

If $d > 0$, the graph shifts **right**

If $c < 0$, the graph shifts **left**

k: horizontal stretch/compression

The graph of $g(x) = f(kx)$ is a horizontal stretch or compression of the graph of $f(x)$ by a factor of $\frac{1}{k}$

If $k > 1$ or $k < -1$, **compressed horizontally** by a factor of $\frac{1}{k}$

If $-1 < k < 1$, **stretched horizontally** by a factor of $\frac{1}{k}$

If $k < 0$, **horizontal reflection** (reflection in the y-axis)

Note: a vertical stretch or compression means that distance from the y-axis of each point of the parent function changes by a factor of $1/k$.

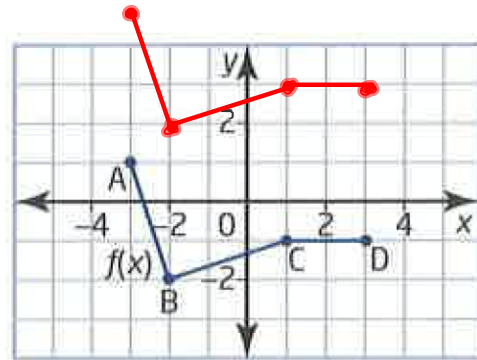
Note: for a horizontal reflection, the point (x, y) becomes point $(-x, y)$

DO IT NOW!

a) Complete the table of values for the function $f(x)$ and $g(x)$.
Then use the table of values to plot image points and graph the function $g(x)$

$f(x) : (x, f(x))$	$g(x) : (x, f(x) + 4)$
A(-3, 1)	A'(-3, 5)
B(-2, -2)	B'(-2, 2)
C(1, -1)	C'(1, 3)
D(3, -1)	D'(3, 3)

↑
base function
↑
image points



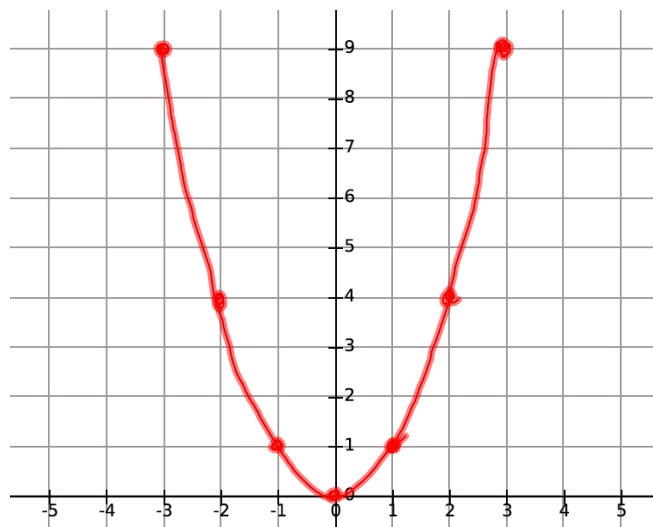
Quadratic Functions

Base Function:

Graph of Base Function

Key Points:

x	y
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9



Order of Transformations

1. stretches, compressions, reflections
2. translations

$$a \rightarrow k \rightarrow d \rightarrow c$$

Example 1: If $f(x) = x^2$, describe the changes and write the transformed function:

a) $g(x) = 2f(x)$

vertical stretch b.a.f.o. 2

$$g(x) = 2x^2$$

b) $g(x) = f(2x)$

horizontal compression
b.a.f.o. $\frac{1}{2}$

$$g(x) = (2x)^2$$

c) $g(x) = f(x) + 4$

shift up 4 units

$$g(x) = x^2 + 4$$

d) $g(x) = f(x+3)$

shift left 3 units

$$g(x) = (x+3)^2$$

e) $g(x) = -f(x)$

vertical reflection
(flip over x -axis)

$$g(x) = -x^2$$

f) $g(x) = f(-x)$

horizontal reflection
(flip over y -axis)

$$g(x) = (-x)^2$$

Example 2: For each of the following functions, describe the transformations to $f(x) = x^2$ in order and write the transformed equation.

a) $g(x) = -2f[-3(x+3)] - 1$ → plug in for x into x^2

vertical stretch by a factor of 2

vertical reflection

horizontal compression by a factor of $1/3$

horizontal reflection

shift left 3 units and down 1 unit

$$g(x) = -2[-3(x+3)]^2 - 1$$

b)

$$y = \frac{1}{2}f[-3(x - 2)] + 5$$

vertical compression by a factor of $\frac{1}{2}$

horizontal compression by a factor of $\frac{1}{3}$

horizontal reflection

shift right 2 units and up 5 units

$$g(x) = \frac{1}{2}[-3(x-2)]^2 + 5$$

Example 3: for each of the following functions...

i) make a table of values for the parent function

ii) graph the parent function $f(x) = x^2$

iii) describe the transformations

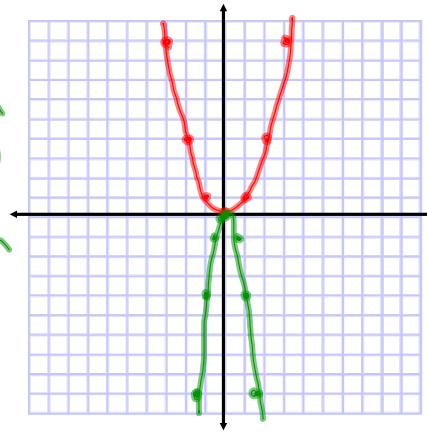
iv) make a table of values of image points

v) graph the transformed function and write its equation

a) $g(x) = -f(2x)$

$a = -1$; vertical reflection
(multiply y-values by -1)

$k = 2$; horizontal compression
by a factor of $1/2$
(divide x-values by 2)



$f(x) = x^2$

x	y
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9

$g(x) = -(2x)^2$

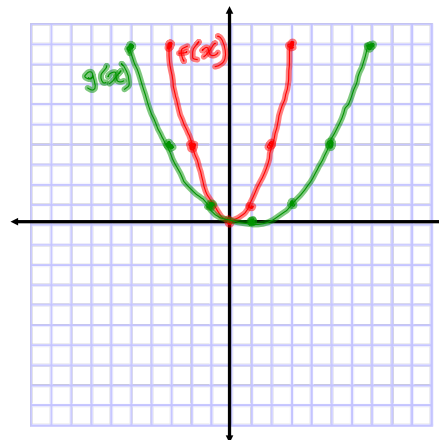
$\frac{x}{2}$	-y
-1.5	-9
-1	-4
-0.5	-1
0	0
0.5	-1
1	-4
1.5	-9

b) $g(x) = f[-1/2(x-1)]$

horizontal stretch by a factor of 2 ($2x$)

horizontal reflection ($-x$)

shift right 1 unit. ($x+1$)



$f(x) = x^2$

x	y
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9

$g(x) = [-\frac{1}{2}(x-1)]^2$

$-2x+1$	y
7	9
5	4
3	1
1	0
-1	1
-3	4
-5	9

c) $g(x) = -2f[-3(x+3)] - 1$

vertical stretch by 2. (2y)

vertical reflection (-y)

horizontal compression by $\frac{1}{3}$ ($\frac{x}{3}$)

horizontal reflection (-x)

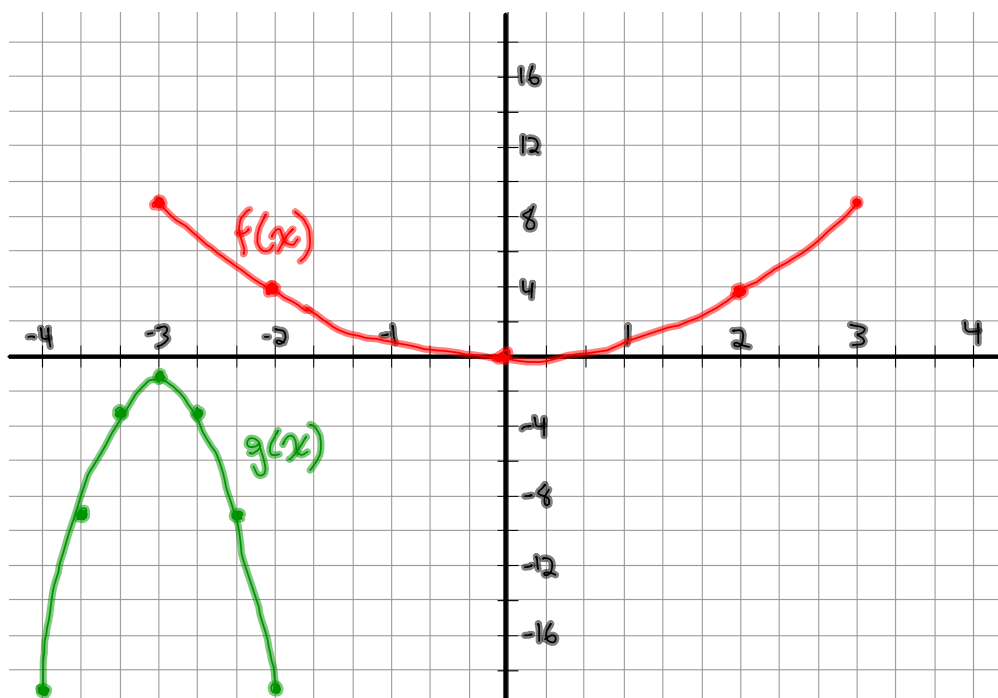
shift left 3 units and down 1. (x-3) (y-1)

$f(x) = x^2$

x	y
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9

$g(x) = -2[-3(x+3)]^2 - 1$

$-\frac{x}{3} - 3$	$-2y - 1$
-2	-19
-2.3	-9
-2.67	-3
-3	-1
-3.3	-3
-3.67	-9
-4	-19



Complete Worksheet